

Autonomous Modular Data Centers for Increased Energy Efficiency and Hardware Longevity

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Abstract—Rapid compute growth is fundamentally constrained by global energy limits. We address this with hermetically sealed Autonomous Modular Data Centers (AMDCs) that maximize compute density per watt. Housed in 40-foot shipping containers, each 1.5 MW module targets an ultra-low PUE of 1.1. This is achieved via an anoxic, 35°C environment using warm water liquid cooling. Simulations show this tech-first approach extends hardware longevity and cuts HVAC costs by up to 56

Index Terms—modular data center, autonomous maintenance, robotics, liquid cooling, stranded power, anoxic environment, PUE, OCP

I. INTRODUCTION

The growth of AI compute is now gated less by silicon than by power. Grid interconnection queues span years, and the most abundant energy sources are often stranded: far from load centers, intermittent, or otherwise hard to monetize. Rather than treat data centers as fixed, human-serviced facilities sited near the grid’s strongest nodes, we treat them as mobile, dispatchable loads that can be delivered to where energy already exists [1]. The idea is not new; containerized modular data centers were proposed as a manufacturable, rapidly deployable alternative to purpose-built halls nearly two decades ago [2].

Every modular data center to date assumes scheduled human access for drive swaps, cable work, and component replacement, and that single assumption tethers the facility to road access, breathable air, and a service workforce. RAY removes it by combining three design choices that reinforce one another: a hermetically sealed anoxic interior, high-temperature warm-water cooling, and an in-situ maintenance robot. The result is a single architecture, the autonomous modular data center (AMDC), that can be sealed at the factory and operated unattended for years.

II. SEALED ANOXIC ENVIRONMENT

The interior of each module is sealed and backfilled with nitrogen to a low-oxygen state. The objective is not merely fire suppression but the elimination of the two dominant drivers of long-term electronic degradation: oxygen and humidity. Oxygen-free preservation at <1% O₂ sustained over decades is well established in museum conservation, where sealed nitrogen enclosures protect sensitive artifacts for the long term [3]. Humidity is equally important: controlled studies of atmospheric corrosion show that suppressing relative humidity is as consequential as suppressing oxygen for preventing corrosion of metallic and electronic surfaces [4]. A dry, anoxic interior therefore directly extends hardware longevity and reduces the failure rate that the maintenance robot must service.

We do not, however, pursue a true inert atmosphere. The cost of oxygen reduction rises steeply as concentration falls, while the marginal corrosion benefit diminishes [5]. Targeting a moderate residual O₂ (on the order of a few percent, comparable to hypoxic fire-prevention systems already deployed in commercial buildings) captures the large majority of the reliability benefit

TABLE I
RAY AMDC REFERENCE DESIGN TARGETS

Parameter	Target
Form factor	40-ft sealed shipping container
IT capacity	1.5 MW
Internal temperature	35 °C
Cooling	Warm-water liquid + free cooling
Target PUE	≈1.1
Atmosphere	Low-O ₂ nitrogen (anoxic)
HVAC savings vs. 25 °C*	up to 56%
Maintenance model	In-situ autonomous robot

*Preliminary simulation; subject to physical validation.

at a fraction of the cost. Off-the-shelf oxygen-reduction and elastomeric sealing technologies are mature enough to hold this state over a multi-year service life, and a purge system provides a fail-safe response to unexpected internal events.

III. HIGH-TEMPERATURE WARM-WATER COOLING

A sealed module without human occupancy removes the constraint that interior temperature be comfortable. We take advantage of this directly: by operating the interior at 35 °C and rejecting heat through warm-water liquid cooling, the facility can rely on free (ambient) cooling across a far wider range of climates, since the higher the supported internal temperature, the larger the fraction of the year that outside air alone suffices for heat rejection [6]. A broad body of work, including U.S. Department of Energy analysis, concludes that operating within wider temperature and humidity envelopes yields substantial operational cost savings without materially compromising IT reliability or availability [5].

Taken together, minimal mechanical chilling, high-efficiency liquid heat transport, and no overhead for human-comfort conditioning target a PUE of approximately 1.1. Our preliminary thermal simulations indicate that raising the internal setpoint from a conventional 25 °C to 35 °C may reduce HVAC energy by up to 56%. These figures are modeled rather than measured and will be validated on the physical prototype, so we report them as design targets. Heat rejection is handled by OCP-standardized coolant distribution units, aligning the module with existing liquid-cooling supply chains rather than bespoke plumbing.

IV. AUTONOMOUS ROBOTIC MAINTENANCE

Sealing the module is only viable if the routine work normally done by technicians can be performed inside, without breaking the atmosphere. There is precedent: robot-managed data centers that mechanically swap and replace server hardware have already been demonstrated commercially [7], and automated maintenance of data-center computers using robotic manipulators is